Lecture 17

Use of Super-Twisting Algorithm for Designing Continuous Controllers

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Sliding mode control and its application

Outline

Motivation

sign(x) as a Disturbance Observer

$$\dot{x} = u + d, \tag{1}$$

where $x \in \mathbb{R}$, $u \in \mathbb{R}$, $d \in \mathbb{R}$. Design $u = -k\text{sign}(x)$ and choose $k > |d|_{\max}$.

It can be shown that when $x = 0$, the equivalent value of control is obtained by setting $\dot{x} = 0$, which implies $d = [k\text{sign}(x)]_{eq}$.

Super Twisting as a Disturbance Observer

$$\begin{aligned} u &= -\lambda|x|^{\frac{1}{2}}\text{sign}(x) + v \\ \dot{v} &= -\alpha\text{sign}(x) \end{aligned} \tag{2}$$

After substituting the control input (2) into (1), one can write

$$\begin{aligned} \dot{x} &= -\lambda|x|^{\frac{1}{2}}\text{sign}(x) + v + d \\ \dot{v} &= -\alpha\text{sign}(x) \end{aligned} \tag{3}$$

Motivation

- Let us define $z := v + d$, then $\dot{z} = \dot{v} + \dot{d}$
- The system can be written as

$$\begin{aligned} \dot{x} &= -\lambda |x|^{\frac{1}{2}} \text{sign}(x) + z \\ \dot{z} &= -\alpha \text{sign}(x) + \dot{d}, \qquad |\dot{d}| \leq \Delta, \end{aligned}$$

  If $\alpha = 1.1\Delta$ and $\lambda = 1.5\sqrt{\Delta}$, it has been proved that $x = z = 0$ in finite time.

- $z = 0$ implies $d = -v$, so one can also construct the disturbance.

The disturbance is given by

$$\begin{aligned} d &= -v \\ \dot{v} &= -\alpha \text{sign}(x) \end{aligned}$$

when x and $v + d$ are zero in (3).

Finite Time Stabilization of Chain of Integrators

Consider the following chain of integrators

$$\begin{aligned} \dot{x}_1 &= x_2 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= u \end{aligned} \tag{4}$$

Theorem: Bhat and Bernstein

Let $k_1, \dots, k_n > 0$ be such that the polynomial $s^n + k_n s^{n-1} + \cdots + k_2 s + k_1$ is Hurwitz, and there exists $\epsilon \in (0, 1)$ such that, for every $\alpha \in (1 - \epsilon, 1)$, the origin is a globally finite time stable equilibrium for the system (4) under the feedback control

$$u = -k_1 |x_1|^{\alpha_1} \text{sign}(x_1) - \cdots - k_n |x_n|^{\alpha_n} \text{sign}(x_n),$$

$$\alpha_{i-1} = \frac{\alpha_i \alpha_{i+1}}{2\alpha_{i+1} - \alpha_i}, \quad i = 2, \dots, n,$$

with $\alpha_{n+1} = 1$ and $\alpha_n = \alpha$.

Finite Time Stabilization of Chain of Integrators with Disturbance

Consider the following uncertain second order system

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u + d, \end{aligned} \tag{5}$$

where $x = [x_1, \ x_2]^T$ is the state vector, $u = u_{\text{nominal}}$ is the control input and d is a matched disturbance. Also assuming $\Delta_0 \geq |d|$ and $\Delta_1 \geq |\dot{d}|$.

- Bhat and Bernstein Control

$$u_{\text{nominal}} = -k_1|x_1|^{\frac{1}{3}}\text{sign}(x_1) - k_2|x_2|^{\frac{1}{2}}\text{sign}(x_2) \tag{6}$$

where $k_1 = 20$ and $k_2 = 9$, and disturbance is $8\sin(t) + 4$ selected for the simulation.

Finite Time Stabilization of Chain of Integrators with Disturbance

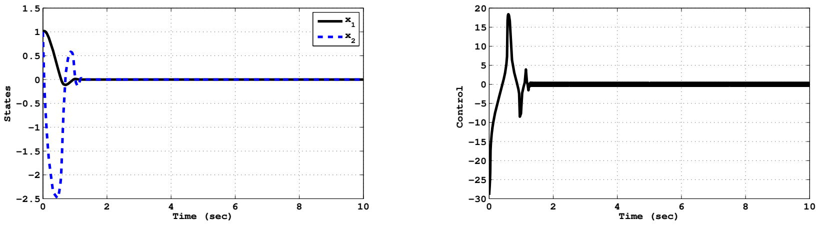

Figure: Finite Time control Bhat et.al., Without Disturbance

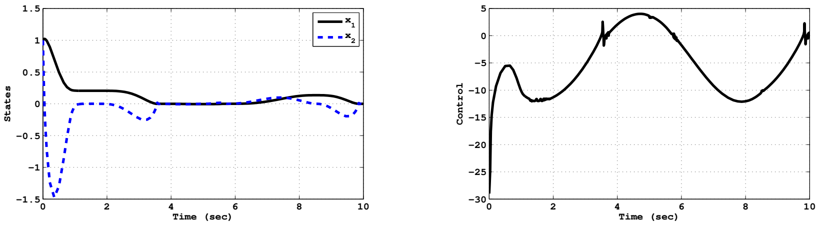

Figure: Finite Time control Bhat et.al., With Disturbance

Integral Sliding Mode Control with Discontinuous Control

- Integral sliding mode control has two parts viz. (i) *nominal control* (u_{nominal}) and (ii) a *discontinuous control* (u_{discon}).
- The nominal control is designed for the system without disturbance to have a desired trajectory. The design of u_{nominal} and u_{discon} are completely independent.
- It may be noted that the equivalent value of the discontinuous control is the negative of the disturbance.
- So, when $u = u_{\text{nominal}} + u_{\text{discon}}$ is applied to the system having disturbance, u_{discon} rejects the disturbance and the desired trajectory is obtained by the application of u_{nominal}.

ISM control input, $u = u_{\text{nominal}} + u_{\text{discon}}$

$$u_{\text{discon}} = -k_3 \text{sign}(s), \qquad k_3 > \Delta_0. \tag{7}$$

- The sliding surface $s \in \mathbb{R}$ is defined as

$$s = x_2 - x_{20} - \int_0^t u_{\text{nominal}} d\tau, \tag{8}$$

Integral Sliding Mode Control with Discontinuous Control

- When the system is on the sliding surface, the equivalent value of the control is calculated by substituting the derivative of the sliding surface equal to zero.
- Hence, mathematically one can write

$$\begin{aligned}\dot{s} &= u + d - u_{\text{nominal}} = 0\\ &= u_{\text{nominal}} + u_{\text{discon}} + d - u_{\text{nominal}} = 0 \Rightarrow u_{\text{discon}} = -d\end{aligned}$$

- Substituting the value of u_{discon} from the (7), one can write that

$$[-k_3 \text{sign}(s)]_{eq} + d = 0$$

- Therefore, when system is on the sliding surface, the value of the disturbance is $d = [k\text{sign}(s)]_{eq}$ and it is canceled out.
- But, it can clearly be observed here that the original control u, which is applied to the plant (5), also contains the discontinuous control.
- Due to the discontinuity in the control chattering phenomenon is generated in actuators, and it is not desirable for the practical implementation.

Proposed Continuous Integral SMC with STC as a Disturbance Observer

- Consider again the same second order system (5), but now the control input for the system is given by $u = u_{\text{nominal}} + u_{\text{STC}}$, where u_{nominal} is nominal control chosen as (6)
- u_{STC} is super twisting control given as

$$\begin{aligned} u_{\text{STC}} &= -k_4|s|^{\frac{1}{2}}\text{sign}(s) + v \\ \dot{v} &= -k_5\text{sign}(s), \end{aligned} \tag{9}$$

- Sliding surface s is defined as (8) and $k_4 = 1.5\sqrt{\Delta_1},\ \ k_5 = 1.1\Delta_1$. Here also the sliding surface is so designed that sliding mode starts from the initial time.
- Differentiating (8) we get $\dot{s} = u + d - u_{\text{nominal}}$
- Substituting the value of u one gets

$$\dot{s} = u_{\text{STC}} + d \tag{10}$$

- After substituting the value of control (9) to (10), one can write

$$\begin{aligned} \dot{s} &= -k_4|s|^{\frac{1}{2}}\text{sign}(s) + v + d \\ \dot{v} &= -k_5\text{sign}(s) \end{aligned} \tag{11}$$

Proposed Continuous Integral SMC with STC as a Disturbance Observer

- Let us define $z = v + d$, so $\dot{z} = \dot{v} + \dot{d}$. After substituting value of z in (11), one can write

$$\begin{aligned} \dot{s} &= -k_4|s|^{\frac{1}{2}}\text{sign}(s) + z \\ \dot{z} &= -k_5\text{sign}(s) + \dot{d} \end{aligned}$$

Consider the following uncertain chain of integrators

$$\begin{aligned} \dot{x}_1 &= x_2 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= u + d \end{aligned} \tag{12}$$

where d is Lipschitz disturbance and $|\dot{d}| \leq d_0$ is bounded.

Proposed Continuous Integral SMC with STC as a Disturbance Observer

Theorem

Let $k_1, \ldots, k_n > 0$ be such that the polynomial $s^n + k_n s^{n-1} + \cdots + k_2 s + k_1$ is Hurwitz, and there exists $k_{n+1}, k_{n+2} > 0$, $\epsilon \in (0,1)$ such that, for every $\alpha \in (1-\epsilon, 1)$, the origin is a globally finite time stable equilibrium for the uncertain system (12) under the feedback control $u = u_{nominal} + u_{STC}$ where

$$u_{nominal} = -k_1|x_1|^{\alpha_1} sign(x_1) - \cdots - k_n|x_n|^{\alpha_n} sign(x_n) \quad and$$

$$\begin{aligned} u_{STC} &= -k_{n+1}|s|^{\frac{1}{2}} sign(s) + v \\ \dot{v} &= -k_{n+2} sign(s), \end{aligned}$$

with the sliding surface $s \in \mathbb{R}$ is proposed as

$$s = x_n - x_{n0} - \int_0^t u_{nominal} d\tau$$

where x_{n0} is the initial value of the state variable x_n and $\alpha_1, \ldots, \alpha_n$ satisfy

$$\alpha_{i-1} = \frac{\alpha_i \alpha_{i+1}}{2\alpha_{i+1} - \alpha_i}, \quad \forall i = 2, \ldots, n,$$

with $\alpha_{n+1} = 1$ and $\alpha_n = \alpha$.

Proposed Continuous Integral SMC with STC as a Disturbance Observer

It is not always possible to convert any system in the form of the uncertain chain of integrators.

Generalized Continuous Integral Like SMC

Consider the system of the following form

$$\dot{x} = Ax + B(u + d) \tag{13}$$

where $x \in \mathbb{R}^{n\times 1}$, $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times 1}$, $u \in \mathbb{R}$, $d \in \mathbb{R}$ are the state, system matrix, input matrix, control and disturbance respectively.

- Control input for the system (13) is given by $u = u_{\text{nominal}} + u_{\text{STC}}$, where u_{nominal} is nominal control and u_{STC} is super twisting control.
- The nominal control u_{nominal} is designed to achieve the specified performance, when system is free from disturbances.
- For example: u_{nominal} can be any linear control, PID, LQR (linear quadratic regulator), state feedback, optimal control, time varying control, adaptive control etc.

Proposed Continuous Integral SMC with STC as a Disturbance Observer

- Required sliding surface for the system (13) is defined as

$$s = G\left[x(t) - x(t_0) - \int_0^t (Ax + Bu_{\text{nominal}})d\tau\right]$$

where $G \in \mathbb{R}^{1\times n}$, $x(t_0)$ are the projection matrix and initial condition of the system respectively.

- Sliding surface is chosen such that the system trajectories start from the sliding surface and if disturbance comes into the picture then u_{STC} becomes active and disturbance becomes compensated.

- Mathematically, above procedure can be explained as

$$\begin{aligned} \dot{s} &= G\left[Ax + Bu + d - Ax - Bu_{\text{nominal}}\right] \\ &= G\left[Ax + B(u_{\text{nominal}} + u_{\text{STC}} + d) - Ax - Bu_{\text{nominal}}\right] \\ &= GB\left[u_{\text{STC}} + d\right] \end{aligned}$$

- Without loss of generality let us consider $GB = 1$ (otherwise u_{STC} control is scaled by $(GB)^{-1}$, one can get

$$\dot{s} = u_{\text{STC}} + d$$

- Further analysis is similar as previous discussions.

STC based On STO

Consider the second order dynamical system

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u + \rho_1 \\ y &= x_1 \end{aligned} \tag{14}$$

y is the output of the system. ρ_1 is a matched disturbance/uncertainties.

- Most of the controller needs all the state information, so first we reconstruct the other state of the system.
- Then we design a super twisting controller based on the estimated information.

Super twisting observer dynamics for system (14)

$$\begin{aligned} \dot{\hat{x}}_1 &= k_1|e_1|^{\frac{1}{2}}\text{sign}(e_1) + \hat{x}_2 \\ \dot{\hat{x}}_2 &= k_2\text{sign}(e_1) + u \end{aligned}$$

Let us define the error $e_1 = x_1 - \hat{x}_1$ and $e_2 = x_2 - \hat{x}_2$.

STC based on STO

Now, the error dynamics is

$$\dot{e}_1 = -k_1|e_1|^{\frac{1}{2}}\text{sign}(e_1) + e_2$$
$$\dot{e}_2 = -k_2\text{sign}(e_1) + \rho_1$$

- It is assumed that $|\rho_1| < \Delta_0$ and Δ_0 is known.
- If we choose $k_1 = 1.5\sqrt{\Delta_0}$ and $k_2 = 1.1\Delta_0$ then error dynamics will goes to zero.
- Once the error e_1 and e_2 is zero, one can say that $x_1 = \hat{x}_1$ and $x_2 = \hat{x}_2$ after finite time $t > T_0$.
- System represented by (14) has relative degree two system with respect to output variable $y = x_1$.
- Therefore one cannot apply the direct STC, because it is applicable for only relative degree one system.
- So we have to define a sliding manifold of the following form to get a relative degree one with respect to sliding manifold.

$$\hat{s} = c_1x_1 + \hat{x}_2 = 0, \quad \text{where } c_1 > 0 \tag{15}$$

STC based On STO

- To synthesize the control law, taking the time derivative of (15)

$$\begin{aligned}\dot{\hat{s}} &= c_1\dot{x}_1 + \dot{\hat{x}}_2 \\ \dot{\hat{s}} &= c_1x_2 + u + k_2\text{sign}(e_1)\end{aligned} \tag{16}$$

- Substituting the $x_2 = e_2 + \hat{x}_2$ in the (16), then we can write as

$$\dot{\hat{s}} = c_1\hat{x}_2 + c_1 e_2 + u + k_2\text{sign}(e_1). \tag{17}$$

- System (14) in the co-ordinate of x_1 and $\hat{s}$ by using (15) and (17),

$$\begin{aligned}\dot{x}_1 &= \hat{s} - c_1x_1 + e_2 \\ \dot{\hat{s}} &= c_1\hat{x}_2 + c_1 e_2 + u + k_2\text{sign}(e_1).\end{aligned} \tag{18}$$

- Now if we will select the control u as

$$u = -c_1\hat{x}_2 - \lambda_1|\hat{s}|^{\frac{1}{2}}\text{sign}(\hat{s}) - \int_0^t \lambda_2\text{sign}(\hat{s})d\tau. \tag{19}$$

where λ_1 and λ_2 are the designed parameters for the controller.

STC based On STO

- Substituting the control input (19) in (18)

$$\dot{x}_1 = \hat{s} - c_1 x_1 + e_2$$

$$\dot{\hat{s}} = c_1 e_2 - \lambda_1 |\hat{s}|^{\frac{1}{2}} \mathrm{sign}(\hat{s}) - \int_0^t \lambda_2 \mathrm{sign}(\hat{s}) d\tau + k_2 \mathrm{sign}(e_1).$$

- The overall closed loop system controller observer together

$$\Pi : \begin{cases} \dot{x}_1 & = \hat{s} - c_1 x_1 + e_2 \\ \dot{\hat{s}} & = c_1 e_2 - \lambda_1 |\hat{s}|^{\frac{1}{2}} \mathrm{sign}(\hat{s}) - \int_0^t \lambda_2 \mathrm{sign}(\hat{s}) d\tau + k_2 \mathrm{sign}(e_1) \end{cases}$$

$$\Xi : \begin{cases} \dot{e}_1 & = -k_1 |e_1|^{\frac{1}{2}} \mathrm{sign}(e_1) + e_2 \\ \dot{e}_2 & = -k_2 \mathrm{sign}(e_1) + \rho_1. \end{cases}$$

- It is already discussed earlier that estimation error of system Ξ converges to zero in finite time, i.e. there exists a $T_0 > 0$ such that for all $t > T_0$, it follows that $e_1 = e_2 = 0$.
- Note that the trajectories of system Π cannot escape to infinity in finite time.
- Usually observer gains are chosen in such a way that observation error converges faster.

STC based On STO

- The closed loop system further we can write as

$$\begin{aligned}\dot{x}_1 &= \hat{s} - c_1 x_1 \\ \dot{\hat{s}} &= -\lambda_1 |\hat{s}|^{\frac{1}{2}} \mathrm{sign}(\hat{s}) - \int_0^t \lambda_2 \mathrm{sign}(\hat{s}) d\tau + k_2 \mathrm{sign}(e_1),\end{aligned} \tag{20}$$

- In another way by adding some new fictitious state variable L we can write

$$\begin{aligned}\dot{x}_1 &= \hat{s} - c_1 x_1 \\ \dot{\hat{s}} &= -\lambda_1 |\hat{s}|^{\frac{1}{2}} \mathrm{sign}(\hat{s}) + L + k_2 \mathrm{sign}(e_1) \\ \dot{L} &= -\lambda_2 \mathrm{sign}(\hat{s})\end{aligned}$$

- It is clear from the above mathematical transformation that, by selecting STO to estimate the state of the second order uncertain system (14) and following the standard way of the STC design as (19) by taking sliding manifold as (15), the second order sliding motion never starts in (20).
- Because $\dot{\hat{s}}$ contains the non-differentiable term $k_2 \mathrm{sign}(e_1)$, which prevents the possibility of lower two subsystem of (20) to act as the super twisting.
- Thus second order sliding motion (so that $\hat{s} = \dot{\hat{s}} = 0$ in finite time) never begins.

STC based On STO

Proposed Strategy

- The aim here, is to design u, such that the second order sliding motion occurs in finite time.
- For this purpose control is selected as

$$u = -c_1\hat{x}_2 - k_2\text{sign}(e_1) - \lambda_1|\hat{s}|^{\frac{1}{2}}\text{sign}(\hat{s}) - \int_0^t \lambda_2\text{sign}(\hat{s})d\tau \tag{21}$$

where, $\lambda_1 > 0$ and $\lambda_2 > 0$ are controller parameter.

Proof

The closed loop system after substituting (21) into (18),

$$\begin{aligned}\dot{x}_1 &= \hat{s} - c_1x_1 + e_2\\ \dot{s} &= c_1e_2 - \lambda_1|\hat{s}|^{\frac{1}{2}}\text{sign}(\hat{s}) - \int_0^t \lambda_2\text{sign}(\hat{s})d\tau\end{aligned} \tag{22}$$

STC based On STO

- As discussed earlier, the observer error converges to zero in finite time, so substituting $e_2 = 0$,

$$\begin{aligned} \dot{x}_1 &= \hat{s} - c_1 x_1 \\ \dot{\hat{s}} &= -\lambda_1 |\hat{s}|^{\frac{1}{2}} \mathrm{sign}(\hat{s}) + \nu \\ \dot{\nu} &= -\lambda_2 \mathrm{sign}(\hat{s}) \end{aligned} \tag{23}$$

where $\nu = -\int_0^t \lambda_2 \mathrm{sign}(\hat{s}) d\tau$.

- The last two equations of (23) have the same structure as super twisting algorithm.
- Therefore, one can easily observe that after finite time $t > T_1$, $\hat{s} = \dot{\hat{s}} = 0$, which further implies, that the closed loop system is given as

$$\begin{aligned} \dot{x}_1 &= -c_1 x_1 \\ x_2 &= -c_1 x_1 \end{aligned}$$

- Therefore, both the states x_1 and x_2 are asymptotically stable by choosing $c_1 > 0$.

Higher-order Sliding Mode Observer based STC

- It is seen that if an n-th order SMO is used to estimate the states of n-th order perturbed integrator system, either the SOSM is not achieved or the controller becomes discontinuous.
- So a continuous controller design is proposed based on $(n+1)$-th order observer.

The higher order sliding mode observer (HOSMO) dynamics for (14)

$$\begin{aligned}\dot{\hat{x}}_1 &= \hat{x}_2 + k_1|e_1|^{\frac{2}{3}}\mathrm{sign}(e_1)\\ \dot{\hat{x}}_2 &= \hat{x}_3 + u + k_2|e_1|^{\frac{1}{3}}\mathrm{sign}(e_1)\\ \dot{\hat{x}}_3 &= k_3\mathrm{sign}(e_1)\end{aligned} \tag{24}$$

Let us define the error as $e_1 = x_1 - \hat{x}_1$, $e_2 = x_2 - \hat{x}_2$.

Higher-order Sliding Mode Observer based STC

If we consider the perturbed n-th order integrators system

$$\begin{aligned} \dot{x}_i &= x_{i+1}, \qquad i = 1, \ldots, n-1 \\ \dot{x}_n &= u + \rho_1 \\ y &= x_1 \end{aligned}$$

the $(n+1)$-th order SMO is given as

HOSMO for perturbed n-th order integrator

$$\begin{aligned} \dot{\hat{x}}_i &= \hat{x}_{i+1} + z_i, \qquad i = 1, \ldots, n-1 \\ \dot{\hat{x}}_n &= \hat{x}_{n+1} + u + z_n \\ \dot{\hat{x}}_{n+1} &= z_{n+1} \end{aligned}$$

where the correction terms are given as

$$\begin{aligned} z_i &= k_i |e_i|^{(n-i+1)/(n+1)} \mathrm{sign}(e_1), \qquad i = 1, \ldots, n-1 \\ z_{n+1} &= k_{n+1} \mathrm{sign}(e_1) \end{aligned}$$

HOSMO based STC

Defining the error variables as $e = x_i - \hat{x}_i$ for $i = 1, \ldots, n$, we have

Error dynamics

$$\begin{aligned}
\dot{e}_1 &= -k_1|e_1|^{\frac{n}{n+1}}\text{sign}(e_1) + e_2 \\
\dot{e}_2 &= -k_2|e_1|^{\frac{n-1}{n+1}}\text{sign}(e_1) - \hat{x}_3 \\
&\vdots \\
\dot{e}_n &= -k_n|e_1|^{\frac{1}{n+1}}\text{sign}(e_1) - \hat{x}_{n+1} + \rho_1 \\
\dot{\hat{x}}_{n+1} &= -k_{n+1}\text{sign}(e_1)
\end{aligned}$$

The error dynamics can be written as

$$\begin{aligned}
\dot{e}_1 &= -k_1|e_1|^{\frac{2}{3}}\text{sign}(e_1) + e_2 \\
\dot{e}_2 &= -k_2|e_1|^{\frac{1}{3}}\text{sign}(e_1) - \hat{x}_3 + \rho_1 \\
\dot{\hat{x}}_3 &= -k_3\text{sign}(e_1)
\end{aligned} \tag{25}$$

HOSMO based STC

- Now define the new variable $e_3 = -\hat{x}_3 + \rho_1$. Also assuming ρ_1 is a Lipschitz and $|\dot{\rho}_1| < \Delta_1$.

$$\begin{aligned}
\dot{e}_1 &= -k_1|e_1|^{\frac{2}{3}}\text{sign}(e_1) + e_2 \\
\dot{e}_2 &= -k_2|e_1|^{\frac{1}{3}}\text{sign}(e_1) + e_3 \\
\dot{e}_3 &= -k_3\text{sign}(e_1) + \dot{\rho}_1
\end{aligned}$$

- The above equation is finite time stable which is already proved in literature.
- So we can conclude that e_1, e_2 and e_3 will converge to zero in finite time $t > T_2$, by selecting the appropriate gains k_1, k_2 and k_3.
- One can find $x_1 = \hat{x}_1$, $x_2 = \hat{x}_2$ and $\hat{x}_3 = \rho_1$ after finite time $t > T_2$.
- To design a super twisting control for a system (14) considering the same sliding surface (15) and taking its time derivative then, one can write expression as

$$\begin{aligned}
\dot{\hat{s}} &= c_1\dot{\hat{x}}_1 + \dot{\hat{x}}_2 \\
\dot{\hat{s}} &= c_1\hat{x}_2 + c_1 e_2 + u + k_2|e_1|^{\frac{1}{3}}\text{sign}(e_1) + \int_0^t k_3\text{sign}(e_1)d\tau
\end{aligned} \tag{26}$$

HOSMO based STC

Representing the system (14) in the co-ordinate of x_1 and $\hat{s}$ by using (15) and (26),

$$\begin{aligned} \dot{x}_1 &= \hat{s} - c_1 x_1 + e_2 \\ \dot{\hat{s}} &= c_1 \hat{x}_2 + c_1 e_2 + u + k_2 |e_1|^{\frac{1}{3}} \text{sign}(e_1) + \int_0^t k_3 \text{sign}(e_1) d\tau \end{aligned} \tag{27}$$

Now the control input u is selected as

$$\begin{aligned} u = -c_1 \hat{x}_2 - k_2 |e_1|^{\frac{1}{3}} \text{sign}(e_1) - \int_0^t k_3 \text{sign}(e_1) d\tau - \lambda_1 |\hat{s}|^{\frac{1}{2}} \text{sign}(\hat{s}) \\ - \int_0^t \lambda_2 \text{sign}(\hat{s}) d\tau \end{aligned} \tag{28}$$

or

$$u = -c_1 \hat{x}_2 - \int_0^t k_3 \text{sign}(e_1) d\tau - \lambda_1 |\hat{s}|^{\frac{1}{2}} \text{sign}(\hat{s}) - \int_0^t \lambda_2 \text{sign}(\hat{s}) d\tau \tag{29}$$

HOSMO based STC

Substituting the control input (28) in the (27),

$$\begin{aligned}\dot{x}_1 &= \hat{s} - c_1 x_1 + e_2\\ \dot{\hat{s}} &= c_1 e_2 - \lambda_1 |\hat{s}|^{\frac{1}{2}} \mathrm{sign}(\hat{s}) + v\\ \dot{v} &= -\lambda_2 \mathrm{sign}(\hat{s})\end{aligned} \tag{30}$$

Now, the overall closed loop system can be represented as

$$\begin{aligned}&\Pi_1 : \begin{cases}\dot{x}_1 &= \hat{s} - c_1 x_1 + e_2\\ \dot{\hat{s}} &= c_1 e_2 - \lambda_1 |\hat{s}|^{\frac{1}{2}} \mathrm{sign}(\hat{s}) + v\\ \dot{v} &= -\lambda_2 \mathrm{sign}(\hat{s}),\end{cases}\\ &\Xi_1 : \begin{cases}\dot{e}_1 &= -k_1 |e_1|^{\frac{2}{3}} \mathrm{sign}(e_1) + e_2\\ \dot{e}_2 &= -k_2 |e_1|^{\frac{1}{3}} \mathrm{sign}(e_1) + e_3\\ \dot{e}_3 &= -k_3 \mathrm{sign}(e_1) + \dot{\rho}_1,\end{cases}\end{aligned} \tag{31}$$

- It is already discussed earlier that estimation error of system Ξ_1 converges to zero in finite time.
- Note that the trajectories of system Π_1 (above) cannot escape to infinity in finite time. So, we can substitute $e_1 = e_2 = 0$.

HOSMO based STC

- Once the error becomes zero, the closed loop system is given by the following expression

$$\begin{aligned}\dot{x}_1 &= \hat{s} - c_1 x_1 \\ \dot{\hat{s}} &= -\lambda_1 |\hat{s}|^{\frac{1}{2}} \text{sign}(\hat{s}) + v \\ \dot{v} &= -\lambda_2 \text{sign}(\hat{s})\end{aligned} \tag{32}$$

- The lower two equation of (32) is super twisting equation, by selecting appropriate gains λ_1 and λ_2, then $\hat{s} = \dot{\hat{s}} = 0$ in finite time. which further implies, that the closed loop system is given as

$$\begin{aligned}\dot{x}_1 &= -c_1 x_1 \\ x_2 &= -c_1 x_1\end{aligned}$$

- Therefore, both the states x_1 and x_2 are asymptotically stable by choosing $c_1 > 0$.
- It is clear from the STC control (28) and (29) expression based on HOSMO (24) is continuous.

HOSMO based STC

- Also, when we design STC control based on HOSMO then one has to tune only the observer gains, according to the first derivative of disturbance, because it is necessary for the convergence of the error variables of the HOSMO.
- However, during controller design there is no explicit gain condition for the λ_2 with respect to the disturbances.
- One can also observe that super twisting output feedback controller (**??**) design based on super twisting observer, requires more conditions on gain relating to disturbance and its derivative.
- One is STO gain means observer gain k_2 based on the explicit maximum bound of the direct disturbance.
- Another is λ_2, STC gain based on the maximum bound of the derivative of disturbance.

HOSMO based STC

- One can conclude from the above observation that sound mathematical analysis reduces the two gains conditions with respect to disturbance by simply one gain condition.
- Also the precision of the sliding manifold is much improved by using the HOSMO based STC rather than STO based STC.
- Due to the increase of this precision of sliding variable precision of the states are also much affected.
- In other word if we talk about stabilization problem, then states are much closer to the origin in the case of HOSMO based STC rather than STO based STC.
- We only talk about the closeness of states variable with respect to an equilibrium point, because only asymptotic stability is possible in both the design methodology.

Thank You!